Supervised Learning for Medical Diagnosis:
A Comparative Analysis of Logistic Regression and Support Vector Machines on a Breast Cancer Dataset

Abstract

Supervised learning now is an elemental method in contemporary data-driven decision-making, especially in high-stakes areas like healthcare. In this tutorial, there is a comparative study of logistic regression and support vectors machine (SVM) when applied to Wisconsin Breast Cancer dataset. The paper will examine the entire supervised learning process, such as preprocessing, feature scaling, model training, and evaluation of the process in terms of accuracy, confusion matrices, ROC curves and others. Specific focus is given to the issue of the impact of model complexity, choice of a kernel, and data characteristics on performance. The findings indicate that although simple linear models can give powerful bases, more flexible models like RBF SVM can be used to identify the delicate trends within the data and enhance the ability to predict. Another vital aspect of using medical machine learning presented in the tutorial is the issue of interpretability, ethics, and accessibility.

1. Introduction

Machine learning has revolutionized the process of analysing complex data in a broad spectrum of fields such as finance to medicine. The supervised learning, as one of the various paradigms of machine learning, has been one of the most popular as it can learn directly using labelled information. In supervised learning, the model is trained with pairs of inputs and outputs, and thus it is able to learn a mapping function which can be employed to forecast results on unknown data (Bishop, 2006).

In classification problems, we are dealing with a task of classifying each input in to a category. This is especially applicable when it comes to medical diagnostics, where the information provided by the patient can be applied to the diagnosis of the condition, whether it is present or not. As an example, with measurements obtained out of medical pictures, a model can be trained to result in the classification of tumours as malignant or benign. These are technically difficult applications, but also have serious implications in the real world, because incorrect predictions may have an impact on patient outcomes.

The current tutorial is dedicated to the application of the supervised learning methods to the problem of breast cancer classification by utilizing the Wisconsin Breast Cancer data. Two popular models are elaborated:

- Logistic Regression, a linear probabilistic classifier which can give interpretable results.

- Support Vector Machine (SVM) is an effective classification algorithm, which can be used to model linear and non-linear relationships.

Additionally, different kernels of SVM (linear, polynomial and radial basis function) are experimented to gain knowledge of the impact of leeway on models. The prime intention of this tutorial is to compare these models not just in numbers but also provide a clear conceptual picture of the behaviour and shortcomings of these models. This is in line with some of the most vital learning outputs in machine learning including the ability to apply algorithms, test performance, and critically analyse outputs.

.2. Dataset Overview

The dataset in this research is the Wisconsin Breast Cancer (Diagnostic) dataset, a commonly known standard in machine learning research (Street et al., 1993). It has a total of 569 observations, which are all tumour samples of a digitised image of the fine needle aspirate of the breast tissue.

Each sample can be characterized by 30 numerical features representing properties of the cell nuclei including:

- Radius (mean distance from centre to boundary)
- Texture (variation in pixel intensity)
- Smoothness
- Concavity
- Symmetry

The characteristics are created to measure minute variations in malignant tumours and benign tumours.

The target variable, which is diagnosing, denotes the tumour to be:

Malignant (M) — cancerous and may cause life-threatening.

Benign (B) — non-cancerous.

This labeled form renders the data perfectly suitable in supervised learning, whereby the model is able to acquire patterns that are related to each category.

The notable attribute of this data is that it is relatively clean and well-organised, therefore the emphasis is made on the model behaviour instead of spending a lot of time cleaning the data. Nevertheless, preprocessing processes that include encoding and scaling are required to achieve top performance.

3. Supervised Learning Pipeline

One of the major features of any machine learning job is how to make a pipeline robust and repeatable. This guarantees reliability of the results and generalisation to new data.

3.1 Label Encoding

Given that the model of machine learning works with numerical data, the categorical labels have to be transformed into numerical numbers. In this case:

- Malignant (M) is encoded as 1
- Benign (B) is encoded as 0

This converts the classification problem into a binary one and the model makes the probabilities of malignancy.

3.2 Train/Test Split

In order to compare the model performance, the dataset will be split into two subsets:

- Training set (80%) - This is applied to train the model.
- Testing set (20%)- to assess performance.

Stratification is used to make sure that subsets of the sample have the same ratio of cases of malignancy and benignity. This is especially relevant in medical data, where the imbalance in the classes may be used to bias the model.

3.3 Feature Scaling

The data in the dataset is highly inconsistent in terms of magnitude. The radius values could be tens, as an example, whilst other features can be significantly smaller. Unless there is scaling, the models like SVM can be biased towards features with higher numerical value.

In response to this, standardization is used such that each feature has:

- Mean = 0
- Standard deviation = 1

SVM particularly is sensitive to feature scaling, since the algorithm is based on distance measurements. It is also effective in enhancing the convergence of optimisation algorithms in the Logistic Regression.

4. Supervised Models

4.1 Logistic Regression

The simplest but the most effective classification algorithms are the Logistic Regression. It is classified, as opposed to regressed, as it is named. The model is an approximation of the probability of a certain input to be in a certain class based on a logistic (sigmoid) function estimation.

The boundary formed by the Logistic Regression is linear, i.e. it tries to separate the classes with a straight line (or a hyper plane in higher dimensions). This helps in its simplicity because the coefficients of each feature can be very easily interpreted to determine their effect on the prediction.

This interpretability is useful especially when it comes to the medical applications. The model becomes more transparent and trustworthy as it allows clinicians to know which characteristics can the most contribute to a diagnosis.

The primary weakness of Logistic Regression is however, that it does not have the ability of capturing complex non-linear relationships unless feature engineering is implemented.

4.2 Support Vector Machines (SVM)

The even more advanced models are known as Support Vector Machines which attempt to locate the best boundary dividing classes with the largest margin possible (Cortes & Vapnik, 1995). The critical point is to optimise the gap between the decision boundary and the nearest data points, the so-called support vectors.

The usability of the kernel functions is one of the strongest features of SVM, and the model processes with the ability of mapping the data into a higher dimensional space in which the data is easier to classify.

There are three kernels discussed in this tutorial:

- Linear Kernel - just like Logistic Regression except that it aims at maximising margin.

- Polynomial Kernel - adds curved boundaries to the decision space.

- RBF (Radial Basis Function) Kernel - very versatile and can be used to model complicated patterns.

There is a significant impact on model performance by the selection of the kernel and therefore it is an important part of designing the SVM.

5. Evaluation Metrics

The accuracy of machine learning models is not the only measure we need to assess to evaluate them, especially in medical practice.
The metrics applied are the following:

- Accuracy - percentage of accurate predictions.

- Precision - the percentage of correct predicted positives.

- Recall - percentage of true positives inaccurately recognized.

- F1 Score - trade off between recall and precision.

- Confusion Matrix - predictions broken down.

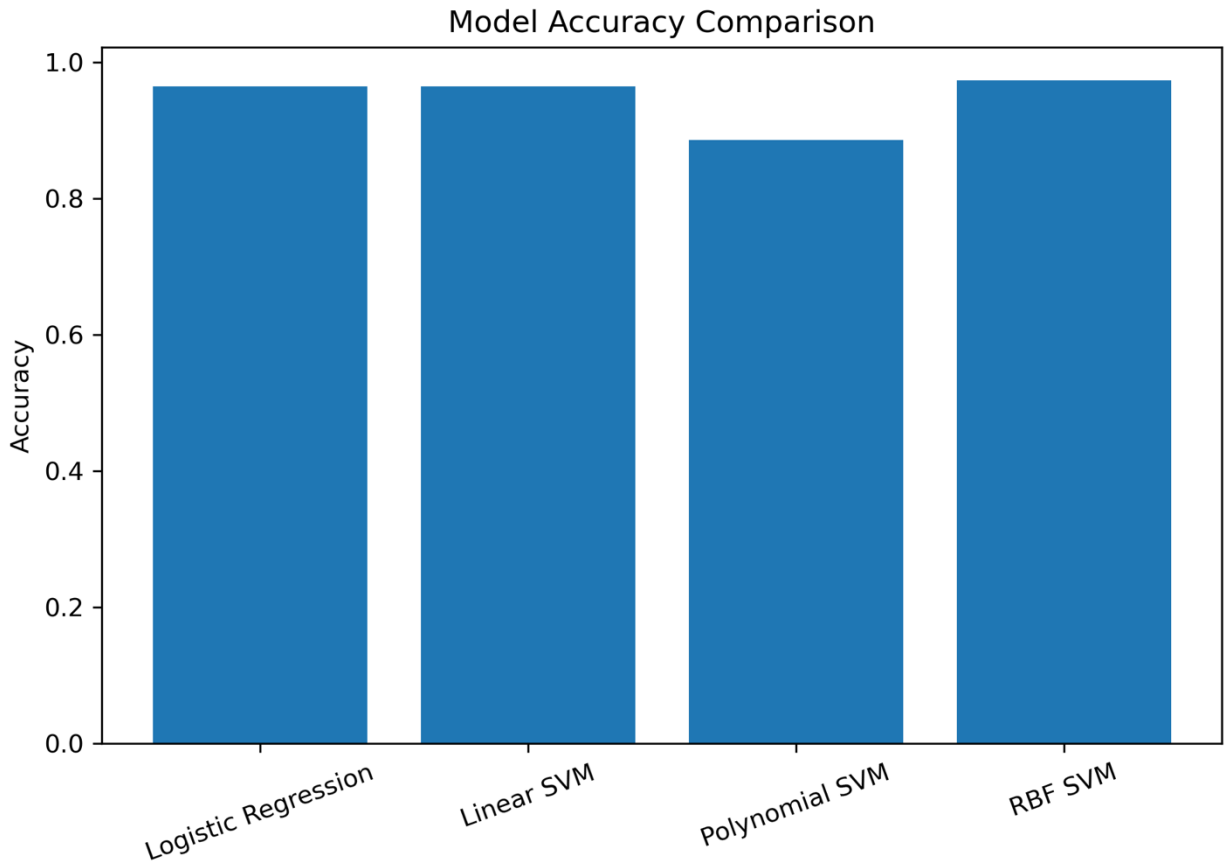
- ROC Curve and AUC - the performance at varying thresholds.

Why These Metrics Matter

In medical diagnosis:
The absence of a malignant tumour (false negative) may result in a delay in treatment.
False positive can lead to unnecessary stress and processes.
Thus, accuracy is not as crucial as recall most of the time.

6. Results and Visualisation

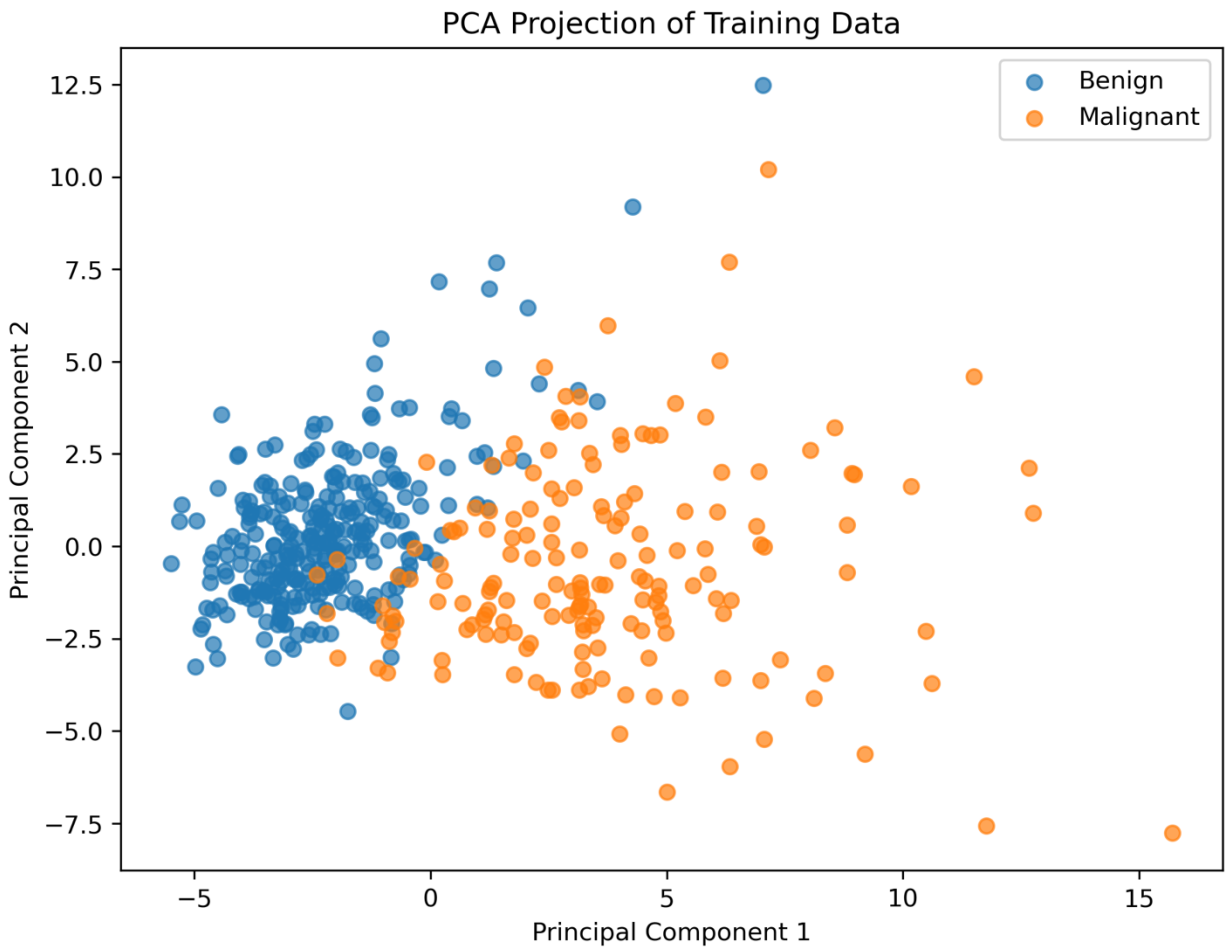
6.1 Accuracy Comparison



As it can be seen, all the models are quite accurate, which means that the dataset is rather well-organized. Nonetheless, RBF SVM is always the top performer followed by the Logistic Regression and Linear SVM.

This implies that the data is mostly linearly separable, but there are non-linear trends that are difficult to capture using linear models that can be modeled by more flexible models.

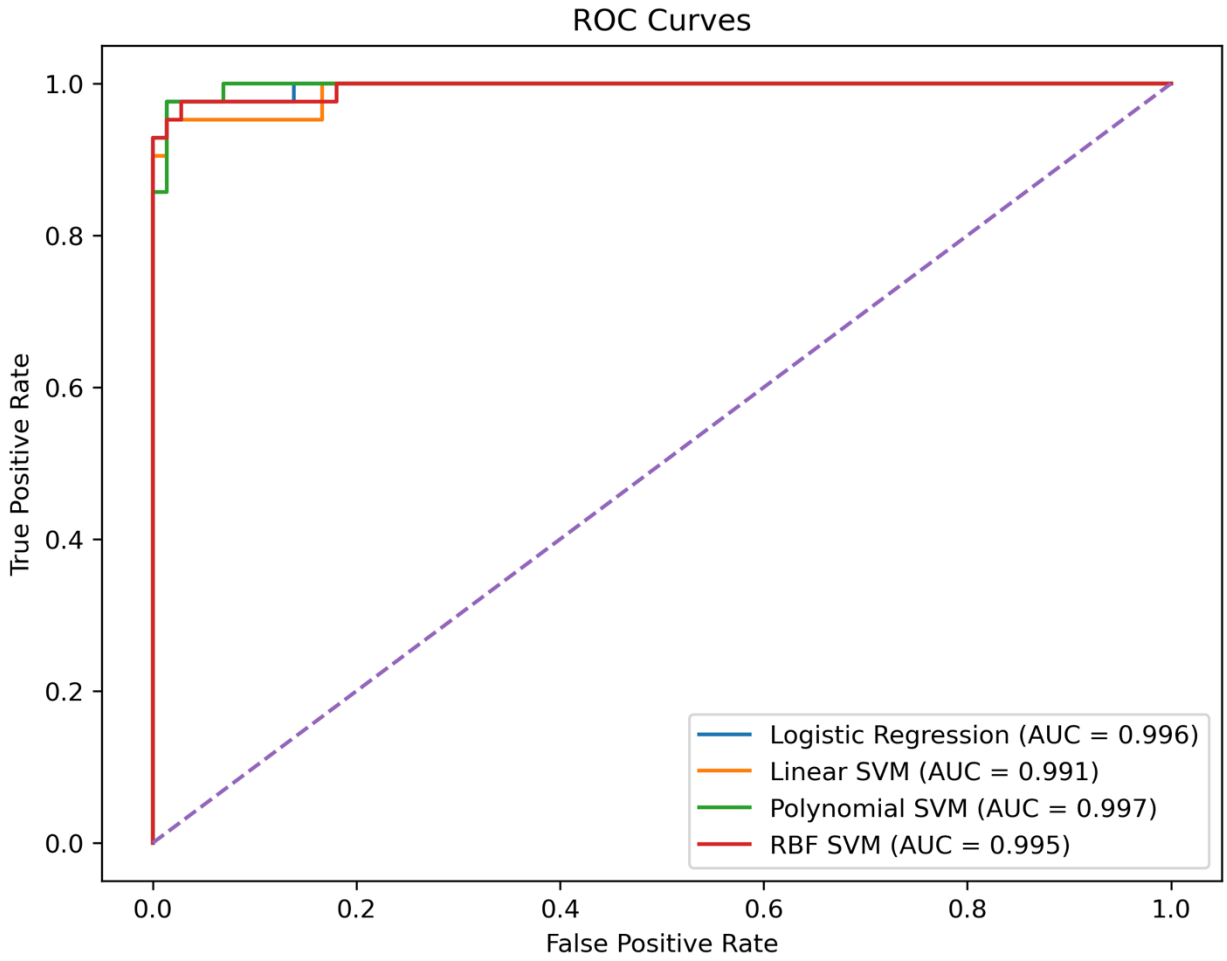
6.2 PCA Projection



The dimensionality of the dataset is reduced in two components with the help of Principal Component Analysis (PCA) to visualise a dataset. The ensuing scatter plot demonstrates that there is partial separation between the benign and malignant samples.

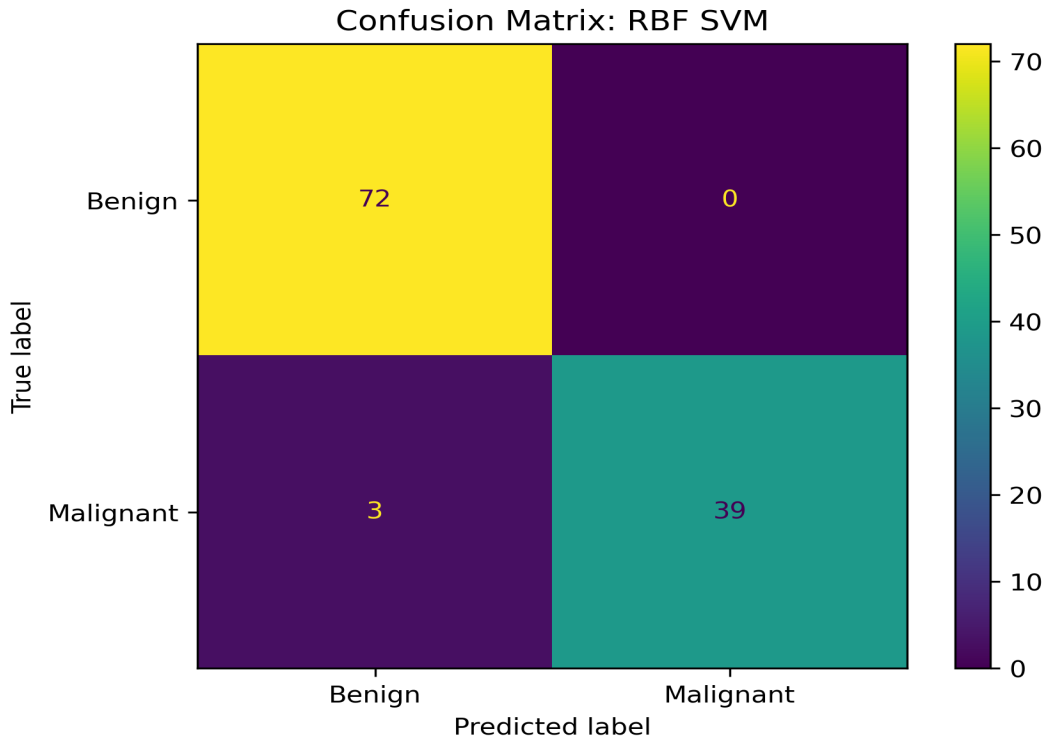
The classes are somewhat overlapping yet visible structure enables supervised models to identify the difference between them.

6.3 ROC Curves



The ROC curves are used to present an overall picture of the model performance under varying classification thresholds. The AUC values are high on all models, but the curve of RBF SVM is nearest to the upper-left corner, which means that it performs better.

6.4 Confusion Matrix



The confusion matrix indicates that the large percentage of samples is correctly classified with minor percentage misclassification. This proves the accuracy of the models especially the RBF SVM.

7. Interpretation of Results

The relevance of models shows some significant insights:

- Logistic Regression is good because the data is almost linear.
- Linear SVM enhances classification, maximising the margin.
- The undue complexity of Polynomial SVM causes overfitting.
- RBF SVM has an optimal balance of flexibility and generalisation.

This is one of the main principles of machine learning:

The complexity of the data should be matched with the model complexity. Classically, to a clinical mindset, models that recall the malignant cases most accurately are desired, since the failure to detect a cancer is very grave.

8. Accessibility Considerations

Responsibility of data communication is important in terms of accessibility. Every visualisation in this tutorial will be designed with:

- Clear labels and titles
- Readable font sizes
- Palettes friendly to colour-blind.

Also, descriptive captions are included so that they can be compatible with screen readers.

9. Ethical Considerations

Machine learning in healthcare has significant ethical issues:

- The problem with bias on training data is potential unfair predictions.
- Absence of transparency of complicated models could decrease trust.
- Making wrong predictions can be a life-threatening problem.

It is thus necessary to make sure that models are well-validated and implemented as a decision-support tool and not a replacement of clinical judgement.

10. Conclusion

This tutorial showed that the supervised learning models could be used to diagnose breast cancer by using the breast cancer dataset.

As indicated by the results, simple models like logistic regression have good and easily understandable base, but more sophisticated models like the RBF SVM have a higher accuracy because they capture complex aspects in the data.

The final decision on the model is determined by the trade-off between accuracy, interpretability and computational complexity. This decision should be made thoroughly in the high-stakes areas like healthcare considering the technical performance as well as ethical aspects.

11. References

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